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Bias Is a Math Problem, AI Bias Is a Technical Problem: 10-Year Literature Review of AI/LLM Bias Research Reveals Narrow [Gender-Centric] Conceptions of `Bias', and Academia-Industry Gap

Sourojit Ghosh, and Kyra Wilson

OBJECTIVE AND METHODS

How do researchers in the field of AI/ML fairness operationalize `bias', and **around what axes of identity they most commonly study it?**

189 papers across 10 years in ***ACL, NeurIPS, FAccT, and AIES** manually screened using questions on defining bias, and proposed implementation

KEY FINDINGS

No definition of `bias' [82% papers]

Negative conceptualization of `bias', conflated with `harm', without considering how `debiasing' is not always optimal

Overrepresentation of Gender Bias Research [79.9% papers]

30.7% = gender-only, 19.9% = gender-occupation, 18% = intersection of gender & race; 73% = binary gender, but 60% acknowledge limitation

Absence of Potential Implementation Plans [89.4% papers]

86.2% papers design `novel' debiasing methods, but do not discuss strategies for implementation into real-world AI systems

High-level Facts

- What aspect(s) of bias is/are mitigated?
- What dataset(s) and model(s) is/are used in experiments?
- How is "success" defined and achieved?
- What methods (qualitative, quantitative, mixed) are used?
- (If human subjects), describe positionality of subjects/annotators.

Limitations and Ethical Considerations

- What are the stated limitations of this research?
- What potential risks can occur if this work is implemented?
- What subpopulation(s) of users is/are the most likely to experience risks?
- How might potential risks cause harm upon affected users?
- What are some mitigation strategies against the potential risks?
- How will strategies differ across affected subpopulations?
- Has this method been tested for other potential amplification of bias(es)?

Intended Use Cases

- In what real-world systems can this work be implemented?
- What real-world AI systems is this work inappropriate for?
- Is the work applicable for a definition of bias within specific cultures and contexts, or can it be applied in global systems?
- In what languages is this tested, and can it be extended?

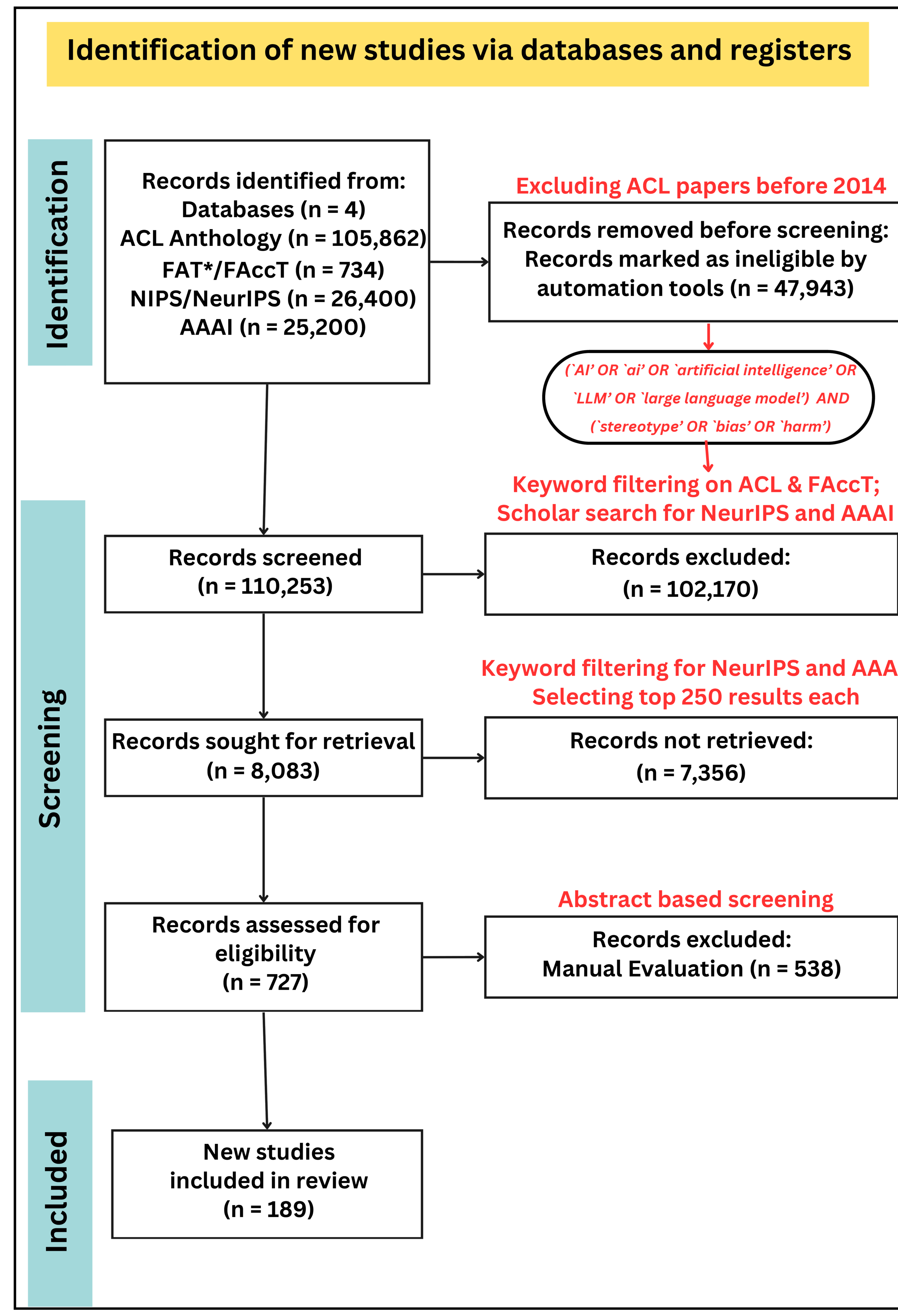
Implementation-level Details

- What computational resources were required to execute this experiment (e.g., GPU hours, memory, latency tradeoffs, etc.)?
- Have experiments addressed performance in multi-agent settings? How did the implementation of this method affect performance?
- Does the method perform debiasing at training- or inference-time?
- Does the implementation of this method require any retraining baseline model(s) from scratch or modifying any internals?
- Are available experiment details and code clearly documented for independent replication of results?

Maintenance and Lifecycle

- Are there likely scenarios where this method might break down?
- Are there maintenance tasks that are required for the long-term success of the implementation of this debiasing method?
- Are there components that can expectedly deprecate over time?
- Does the system support human-in-the-loop extensions post-deployment?

Contact <researcher name> at <email> for more information!



TAKEAWAYS

`Bias' == `bad' promotes purely technical approaches to `debiasing'

Overstudying gender bias can cause harm to other marginalized populations

Lack of implementation plans widens `Academia-Industry' Gap

RECOMMENDATIONS

Define and Expand focus of `Bias' Coverage

Discuss Implementation Plans via System Fact Sheets

Preprint of paper